

MEE342 - Project Calculations

Team: Shift Happens

Date: 3/14/2026

format longG

Material Properties

Dimensions and Input Values

```
D = []; % Diameter of both input and output rods
% NOTE: Gear Notation/Naming Convention:
% "Gear" = G, "Diameter" = D, "Input shaft" = I, "Output shaft" = O,
% "Number of teeth" = N
% 1:6 references the gears from closest to the gear shifter (1) to the furthest
% from the gear shifter (6).

% Gear One
DG1I = 181.7767E-3; % Diameter of Gear closest to the gear shifter
on the input rod
DG1O = 75.7107E-3; % Diameter of Gear closest to the gear shifter
on the output rod
NG1I = 50; % Number of Teeth of Gear closest to the gear
shifter on the input rod
NG1O = 20; % Number of Teeth of Gear closest to the gear
shifter on the output rod

% Gear Two
DG2I = 153.4924E-3;
DG2O = 103.9949E-3;
NG2I = 42;
NG2O = 28;

% Gear Three
DG3I = 139.3503E-3;
DG3O = 118.1371E-3;
NG3I = 38;
NG3O = 32;

% Gear Four
DG4I = 132.2792E-3;
DG4O = 125.2082E-3;
NG4I = 36;
NG4O = 34;
```

% Gear Five

```
DG5I = 128.7437E-3;  
DG5O = 128.7437E-3;  
NG5I = 35;  
NG5O = 35;
```

% Gear Six

```
DG6I = 118.1371E-3;  
DG6O = 139.3503E-3;  
NG6I = 32;  
NG6O = 38;
```

```
T_I = 1500; % [N.m]Torque from the Input rod that the gears  
will be fixed to utilizing the synchros.  
eff = 1; % Assumed efficiency of the gear is 100%
```

Critical Point Calculations

Gear Ratios

G1 = NG10/NG1I

G1 = 0.4

G2 = NG20/NG2I

G2 = 0.6666666666666667

G3 = NG30/NG3I

G3 = 0.842105263157895

G4 = NG40/NG4I

G4 = 0.9444444444444444

G5 = NG50/NG5I

G5 = 1

G6 = NG60/NG6I

G6 = 1.1875

Torque in Input/Output Shaft

T_10 = T_I * (NG10/NG1I) * eff % Torque Output by the first gear

T_10 =

600

```
T_20 = T_I * (NG20/NG2I) * eff      % Torque Output by the second gear
```

```
T_20 =  
1000
```

```
T_30 = T_I * (NG30/NG3I) * eff      % Torque Output by the third gear
```

```
T_30 =  
1263.15789473684
```

```
T_40 = T_I * (NG40/NG4I) * eff      % Torque Output by the fourth gear
```

```
T_40 =  
1416.66666666667
```

```
T_50 = T_I * (NG50/NG5I) * eff      % Torque Output by the fifth gear
```

```
T_50 =  
1500
```

```
T_60 = T_I * (NG60/NG6I) * eff      % Torque Output by the sixth gear
```

```
T_60 =  
1781.25
```

References for the Gear calculations:

<https://gearsolutions.com/features/determination-of-the-agma-j-factor-for-internal-spur-gears/>

https://www.engineersedge.com/calculators/agma_gear_tooth_bending_stress_15856.htm

Shear Stress on Gear Teeth

```
% AGMA Bending Stress Calculation Done using the highest stress gear ratio  
  
% INPUT PARAMETERS  
% Power & Speed  
P = 10e3;          % Power (W)  
EngineRPM = 2000;  
OverDriveRatiotoFinalDriveRatio = G6;          % Highest Gear ratio so  
that we are calculating based off of maximum torque applied  
n = (EngineRPM * OverDriveRatiotoFinalDriveRatio); % Pinion speed (rpm)  
  
% Gear Geometry  
N = NG6I;          % Number of teeth  
m = 2.5;           % Module (m)  
phi = 14.5;        % Pressure angle (deg)  
F = 0.025;         % Face width (m)
```

```

% AGMA Factors (user-defined or from charts)
Ko = 1.5; % Overload factor : Assuming
mean of shock
Ks = 1.0; % Size factor : Assuming
smaller gears for our specific assembly
C_pm = 1.0; % Straddle Mounted gears
pd = 50; % Assumed Pitch diameter is
equivalent to nominal diameter
C_pf = (F/(10*pd)) - 0.025; % Pinion Proportion Factor
C_ma = 1; % Assumed perfect manufacturing
and installation
C_mc = 1.0; % The teeth are not crowned/
lead corrected
C_e = 0.8; % Assumed mesh alignment
correction
Km = 1 + C_mc*(C_pf*C_pm + C_ma*C_e) ; % Load distribution factor
Kb = 1.5; % Rim thickness factor for
Solid Gears
LewisFormFactor = 0.154 - (0.912/N);
Kf = 1.75; % Stress Concentration Factor
J = LewisFormFactor/Kf; % Geometry factor

% Dynamic factor constants (AGMA approximation)
Qv = 8; % Quality number (typical: 6-11)

% STEP 1: PITCH DIAMETER
d = m * N; % Pitch diameter (m)

% STEP 2: TORQUE
omega = (2*pi*n)/60; % Angular velocity (rad/s)
T = P / omega; % Torque (N.m)

% STEP 3: TANGENTIAL LOAD
Wt = (2 * T) / d; % Tangential force (N)

% STEP 4: PITCH LINE VELOCITY
V = (pi * d * n) / 60; % Pitch line velocity (m/s)

% STEP 5: DYNAMIC FACTOR Kv
B = 0.25 * (12 - Qv)^(2/3);
A = 50 + 56 * (1 - B);
Kv = ((A + sqrt(V)) / A)^B;

% STEP 6: AGMA BENDING STRESS
sigma = (Wt * Ko * Kv * Ks * Km * Kb) / (F * m * J);

%% ----- OUTPUT -----
fprintf('Pitch Diameter (m): %.4f\n', d);

```

Pitch Diameter (m): 80.0000

```
fprintf('Torque (N·m): %.2f\n', T);
```

Torque (N·m): 40.21

```
fprintf('Tangential Load Wt (N): %.2f\n', Wt);
```

Tangential Load Wt (N): 1.01

```
fprintf('Pitch Line Velocity (m/s): %.2f\n', V);
```

Pitch Line Velocity (m/s): 9948.38

```
fprintf('Dynamic Factor Kv: %.3f\n', Kv);
```

Dynamic Factor Kv: 1.741

```
fprintf('AGMA Bending Stress (MPa): %.6f\n', sigma*1E-6);
```

AGMA Bending Stress (MPa): 0.001559